01

Lecture 01

GSM, GPRS and EDGE

جی‌اس‌ام، جی‌پی‌آر‌اس و اج

2G — 3G — 4G — 5G — 6G

COURSE PART
Part I – 2G/3G foundations (Lectures 1–4)

ESTIMATED STUDY TIME
6–8 hours (3 h reading, 3–5 h exercises)

DIFFICULTY
Introductory to intermediate

PREREQUISITES
Decibels, basic digital modulation, TDMA/FDMA, the OSI model

In this lecture

1 Learning objectives . . . 2
2 Prerequisite knowledge . . . 2
3 Introduction and motivation . . . 3
4 Fundamental concepts . . . 4
5 Detailed technical explanation . . . 5
6 Mathematical models . . . 7
7 Network architecture and diagrams . . . 10
8 Step-by-step procedures . . . 11
9 Worked examples . . . 12
10 Technology comparison . . . 14
11 Key takeaways . . . 14
12 Exercises . . . 15
13 Further reading . . . 17

Original lecture notes for the course *Advanced Communication Networks*. The lecture sequence follows the course syllabus (26 lectures, 2G to 6G). Equations use identical notation in the English and Persian editions. Companion edition: Persian (`lecture-01-fa.pdf`).

1 Learning objectives

After studying this lecture you should be able to:

- **LO1** **Draw** the GSM reference architecture, name every network element (MS, BTS, BSC, MSC, VLR, GMSC, HLR, AuC, EIR) and state the interface between each pair (Um, Abis, A, C, D, E, F).
- **LO2** **Derive** the GSM radio timing (bit period, burst, TDMA frame and 26-multiframe) from the 270.833 kbit/s gross rate and **compute** the 22.8 kbit/s gross rate of a full-rate traffic channel.
- **LO3** **Explain** the GSM challenge–response authentication with the triplet (RAND, SRES, K_c) and identify its security weaknesses.
- **LO4** **Describe** the GPRS overlay (PCU, SGSN, GGSN, Gb, Gn, Gi, Gr) and the PDP context, and **trace** a PDP context activation step by step.
- **LO5** **Calculate** per-user GPRS/EDGE throughput from the coding scheme and the number of timeslots, and **evaluate** the trade-off between robustness and rate.
- **LO6** **Relate** cluster size, path-loss exponent and co-channel interference through the first-tier C/I model.

2 Prerequisite knowledge

The lecture assumes the decibel arithmetic of the prerequisites review ($x_{\mathrm{dB}} = 10\log_{10} x$), the notion of a modulation symbol and of bandwidth, and the difference between frequency-division (FDMA) and time-division (TDMA) multiple access. If you are unsure about any of these, revise them before continuing; every quantitative result below is built from them.

Key terms. The table lists the vocabulary used throughout the lecture with its Persian equivalent, so that both editions can be read side by side.

Term	Persian	Meaning in this lecture
Mobile Station (MS)	ایستگاه موبایل	Handset = mobile equipment (ME) + SIM card.
Base Transceiver Station (BTS)	ایستگاه فرستنده‌گیرندهٔ پایه	Radio site: transmitters, receivers and antennas of one or more cells.
Base Station Controller (BSC)	کنترل‌گر ایستگاه پایه	Manages radio resources and handovers of many BTSs.
Mobile Switching Centre (MSC)	مرکز سوئیچ موبایل	Circuit switch for calls; also runs mobility signalling.
Home Location Register (HLR)	ثبات مکان خانگی	Permanent subscriber database of the home operator.
Visitor Location Register (VLR)	ثبات مکان مهمان	Temporary copy of data for subscribers in an MSC area.
Authentication Centre (AuC)	مرکز احراز هویت	Stores the secret key K_i and generates triplets.
Timeslot (TS)	شیار زمانی	One of the eight TDMA positions on a carrier.
Burst	بُرست	The radio waveform transmitted in one timeslot.
PDP context	زمینهٔ PDP	Packet session state: address, APN, QoS, tunnel IDs.
SGSN / GGSN	گره‌های پشتیبان GPRS	Serving and gateway packet nodes of the GPRS core.
Coding scheme (CS/MCS)	طرح کدگذاری	Combination of modulation and channel-code rate.

3 Introduction and motivation

In the 1980s Western Europe operated several incompatible analogue cellular systems. A handset bought in one country was useless across the border, every system needed its own equipment industry, and analogue FM voice offered no protection against eavesdropping. The engineering problem that GSM[1] set out to solve was therefore not only a radio problem; it was a *system* problem:

- **Interoperability.** Standardised, open interfaces would let an operator buy base stations from one vendor and switches from another.
- **Roaming.** A subscriber identity independent of the handset (the SIM) and a home database reachable from any visited network.
- **Spectral efficiency and capacity.** Digital speech coding, channel coding and TDMA let several users share one carrier and tolerate more interference, so frequencies can be reused more

[1] Originally *Groupe Spécial Mobile*, later *Global System for Mobile communications*.

tightly.

- **Security.** Authentication of the subscriber and encryption of the radio link.

GSM was designed for circuit-switched voice. When data demand grew in the 1990s, the same radio network was extended with a packet-switched overlay: *GPRS* (General Packet Radio Service), and later *EDGE* (Enhanced Data rates for GSM Evolution), which raised the bit rate per timeslot with a higher-order modulation. Studying these three steps in one lecture shows a pattern that recurs throughout the course: **each generation reuses what it can and adds the minimum new machinery needed for the new problem.**

4 Fundamental concepts

4.1 Cells, frequency reuse and multiple access

A mobile network divides its service area into *cells*, each served by one set of antennas. Because spectrum is scarce, the same carrier frequencies are *reused* in cells that are far enough apart. GSM combines two multiple-access methods:

- **FDMA:** the band is divided into carriers spaced 200 kHz apart. In the primary GSM-900 band, the uplink occupies 890 MHz–915 MHz and the downlink 935 MHz–960 MHz, giving 124 usable carrier pairs with a duplex spacing of 45 MHz [2].
- **TDMA:** each carrier is divided in time into frames of eight timeslots; a full-rate voice user occupies one timeslot in every frame.

Definition 1 • Physical and logical channel

A *physical channel* is a timeslot number on a carrier (or a hopping sequence of carriers). A *logical channel* is a type of information — traffic (TCH) or control (BCCH, CCCH, SDCCH, SACCH, FACCH) — that is mapped onto a physical channel according to a fixed multiframe structure.

4.2 Separating identity from equipment

GSM separates the *subscriber* from the *device*. The subscriber is identified by the IMSI (International Mobile Subscriber Identity: mobile country code, mobile network code and a subscriber number, at most 15 digits) stored on the SIM together with a 128-bit secret key K_i. The device is identified by its IMEI. To avoid sending the IMSI over the air, the VLR assigns a temporary identity, the TMSI, which is used instead whenever possible [6].

4.3 Circuit switching versus packet switching

Voice in GSM is *circuit switched*: when a call is set up, a timeslot on the radio, a channel on the Abis and A links and a path through the MSC are reserved for the whole call, whether or not anybody is speaking. Data in GPRS is *packet switched*: radio blocks are assigned only when the user has data to send, and many users share the same timeslots. This difference is the reason GPRS needs new nodes rather than just new software in the MSC, and it is analysed in depth in Lecture 2.

5 Detailed technical explanation

5.1 The GSM network elements

Mobile Station (MS). The MS is the mobile equipment plus the SIM. The SIM runs the authentication algorithm A3 and the key-generation algorithm A8 so that K_i never leaves the card.

Base Transceiver Station (BTS). The BTS contains the radio transceivers (TRX) of one or more cells. It performs modulation (GMSK), channel coding, interleaving, encryption with A5 and the radio measurements that feed handover decisions. It is intentionally simple: decisions are taken by the BSC.

Base Station Controller (BSC). One BSC controls tens to hundreds of BTSs. It allocates radio channels, manages power control and decides most handovers between its own cells. Together the BTSs and their BSC form the *Base Station Subsystem* (BSS). Speech transcoding between the 13 kbit/s radio codec and the 64 kbit/s PCM format of the core is performed in a transcoder unit (TRAU), usually located at the MSC site to save transmission capacity.

Mobile Switching Centre (MSC) and VLR. The MSC is a telephone exchange extended with mobility functions: it sets up and releases calls, controls inter-BSC handovers, and runs location updating and authentication. The VLR (in practice integrated in the MSC) caches the subscription data and security triplets of every subscriber currently in its area.

Gateway MSC (GMSC). Calls arriving from the fixed network (PSTN/ISDN) are routed to a GMSC, which interrogates the HLR to find where the called subscriber is.

HLR, AuC and EIR. The HLR stores the permanent subscription profile and the address of the VLR currently serving the subscriber. The AuC holds K_i and computes authentication triplets. The EIR holds lists of white, grey and black-listed IMEIs.

5.2 Interfaces

Table 1. Main GSM interfaces and their protocols [5].

Interface	Between	Main protocol	Role
Um	MS – BTS	LAPDm, RR, MM, CC	Radio interface, TDMA/FDMA, GMSK
Abis	BTS – BSC	LAPD over E1/T1	Transport of traffic and BTS control
A	BSC – MSC	BSSAP over SS7	Call and mobility signalling, 64 kbit/s circuits
B	MSC – VLR	MAP	Internal in practice (MSC/VLR combined)
C	GMSC – HLR	MAP	Routing interrogation for incoming calls
D	VLR – HLR	MAP	Location updating, subscriber data
E	MSC – MSC	MAP + ISUP	Inter-MSC handover, call routing
F	MSC – EIR	MAP	IMEI check

The value of an open interface is economic as well as technical: because the A interface is fully standardised, an operator can connect any vendor's BSS to any vendor's MSC. The Abis interface was standardised too, but in practice it often contains vendor-specific operation and maintenance

extensions.

5.3 The radio interface Um

GSM uses Gaussian Minimum Shift Keying (GMSK) with a bandwidth–time product of $BT = 0.3$. GMSK is a constant-envelope modulation, so the handset power amplifier can run near saturation, which saves battery. One bit is carried per symbol, at a gross rate of $13/48\,\text{MHz} \approx 270.833\,\text{kbit/s}$. The time structure derived from this rate is drawn in Figure 4 and computed in Section 6.

A normal burst carries two blocks of 57 data bits, a 26-bit training sequence in the middle (the receiver uses it to estimate the multipath channel for its equaliser), two stealing flags, tail bits and a guard period of 8.25 bit durations that absorbs timing uncertainty and power ramping. Frequency hopping (up to 217 hops per second, one hop per TDMA frame) averages fading and interference across carriers.

5.4 Mobility: location areas and the TMSI

Cells are grouped into *location areas* (LAs). An idle MS listens to the broadcast channel and performs a *location update* only when it enters a new LA, or periodically. When a call arrives, the network pages the MS in all cells of its current LA. The size of an LA is a trade-off between update signalling (small LAs) and paging load (large LAs); this trade-off is quantified in Lectures 4 and 12.

5.5 Security: authentication and ciphering

GSM authentication is a *challenge–response* protocol [7]. The AuC draws a 128-bit random number RAND and computes

$$\text{SRES} = \text{A3}(K_i, \text{RAND}), \qquad (32\ \text{bits}) \tag{1a}$$

$$K_c = \text{A8}(K_i, \text{RAND}), \qquad (64\ \text{bits}) \tag{1b}$$

and sends the triplet $(\text{RAND}, \text{SRES}, K_c)$ to the VLR. The VLR sends only RAND to the MS; the SIM repeats the computation of Eq. (1a) and returns its result. If it matches SRES, the subscriber is authentic and both sides now share K_c, which keys the A5 stream cipher on the radio link. Note what this design achieves and what it does not:

- K_i never travels over any interface, and RAND is useless without K_i.
- Authentication is *one-way*: the network is not authenticated to the MS, so a false base station can impersonate the network.
- Encryption ends at the BTS; Abis, A and the core links are not protected by A5.
- K_c has only 64 bits, and the A5/1 and A5/2 ciphers have published cryptanalytic attacks; A5/2 has been withdrawn and A5/3 (based on KASUMI) was added later.

UMTS (Lecture 3) and LTE (Lectures 9 and 11) fix these weaknesses with mutual authentication, integrity protection and longer keys.

5.6 GPRS: a packet overlay

GPRS adds packet switching while reusing the GSM radio carriers, sites and subscriber databases [1]. Three new functions are needed:

- **Packet Control Unit (PCU)** in the BSS, which implements the packet-oriented RLC/MAC layer and schedules radio blocks among users.
- **Serving GPRS Support Node (SGSN)**, the packet counterpart of the MSC: mobility management (routing areas), authentication, ciphering (in GPRS, ciphering runs between MS and SGSN at the LLC layer) and charging.
- **Gateway GPRS Support Node (GGSN)**, the border router to external packet data networks: it allocates the user IP address and anchors the session.

The SGSN connects to the BSS over *Gb* (originally Frame Relay), to the GGSN over *Gn* using the GPRS Tunnelling Protocol (GTP), and to the HLR over *Gr* using MAP. The GGSN connects to the Internet or a corporate network over *Gi*. The idea of a *tunnel* between a serving node and a gateway node is the single most durable concept introduced by GPRS: it survives as the S5/S8 interface of LTE and the N9 interface of 5G.

On the radio side, GPRS defines packet data channels (PDCH). A PDCH is an ordinary timeslot dedicated to packet traffic; data is sent in *radio blocks* of four consecutive bursts (one block every 20 ms on average). Four coding schemes trade protection for rate: CS-1 (9.05 kbit/s), CS-2 (13.4), CS-3 (15.6) and CS-4 (21.4 kbit/s) per timeslot. A multislot-capable MS can use several timeslots at once.

5.7 GPRS mobility management states

A GPRS MS is in one of three mobility-management states: **IDLE** (not attached; the network does not know the MS), **STANDBY** (attached; location known per routing area; must be paged before data can be delivered) and **READY** (location known per cell; data can be sent immediately). A timer moves the MS from READY back to STANDBY after a period of inactivity. The same idea — keep the device reachable cheaply while idle, and spend signalling only when traffic flows — reappears as ECM-IDLE/CONNECTED in LTE and CM-IDLE/CONNECTED in 5G.

5.8 EDGE: higher-order modulation on the same carrier

EDGE keeps the GSM symbol rate (270.833 ksymbol/s), the 200 kHz carrier and the TDMA structure, but introduces 8-PSK, which carries three bits per symbol instead of one. Enhanced GPRS (EGPRS) defines nine modulation and coding schemes: MCS-1 to MCS-4 use GMSK and MCS-5 to MCS-9 use 8-PSK, from 8.8 kbit/s up to 59.2 kbit/s per timeslot. Two mechanisms make the higher rates usable:

- **Link adaptation:** the network selects the MCS from channel-quality measurements, moving to a robust scheme when the user is at the cell edge.
- **Incremental redundancy:** if a block is received in error, the retransmission carries different puncturing of the same coded bits, and the receiver combines both copies. This is an early form of the hybrid ARQ used in HSPA, LTE and NR.

6 Mathematical models

6.1 GSM radio timing

Engineering problem. Every timing parameter of GSM — the burst length, the frame duration, the maximum cell radius — follows from one number, the gross bit rate. We derive them to see where the familiar values come from.

System model and assumptions. One carrier, GMSK with one bit per symbol, gross rate $R = f_{\text{ref}}/48$ with reference clock $f_{\text{ref}} = 13\,\text{MHz}$. A timeslot lasts 156.25 bit periods and a TDMA frame contains eight timeslots.

Derivation. The bit period and the derived durations are

$$T_{\text{b}} = \frac{1}{R} = \frac{48}{13 \times 10^6}\ \text{s} \approx 3.692\,\mu\text{s}, \tag{2}$$

$$T_{\text{slot}} = 156.25\,T_{\text{b}} = \frac{15}{26}\ \text{ms} \approx 576.9\,\mu\text{s}, \tag{3}$$

$$T_{\text{frame}} = 8\,T_{\text{slot}} = \frac{60}{13}\ \text{ms} \approx 4.615\,\text{ms}, \tag{4}$$

$$T_{26} = 26\,T_{\text{frame}} = 120\,\text{ms}. \tag{5}$$

Gross rate of a full-rate traffic channel. In each 26-multiframe a full-rate TCH uses 24 frames for traffic, one for the SACCH and leaves one idle. Each traffic burst carries $2 \times 57 = 114$ coded bits, so

$$R_{\text{TCH/F}} = \frac{114\ \text{bit} \times 24}{T_{26}} = \frac{2736\ \text{bit}}{0.120\ \text{s}} = 22.8\,\text{kbit/s}. \tag{6}$$

The 13 kbit/s speech codec output is protected by channel coding that expands it to exactly this 22.8 kbit/s.

Physical interpretation. The ratio $R_{\text{TCH/F}}/R = 22.8/270.833 \approx 8.4\%$ is close to $1/8$ minus the share of the frame spent on training sequence, tail bits, guard time and signalling. GSM therefore spends roughly a quarter of its gross capacity on overhead — a figure to remember when comparing it with OFDM systems in Lectures 6 and 14.

Spectral efficiency. With $B = 200\,\text{kHz}$ the gross spectral efficiency is $\eta = R/B \approx 1.35\,\text{bit/s/Hz}$. The Shannon bound [10]

$$C = B \log_2(1 + \text{SNR}) \tag{7}$$

with C the capacity in bit/s, B the bandwidth in Hz and SNR the linear signal-to-noise ratio, shows that 1.35 bit/s/Hz could in principle be carried at $\text{SNR} = 2^{1.354} - 1 \approx 1.56$ ($\approx 1.9\,\text{dB}$). GSM operates at a far higher SNR (around 9 dB carrier-to-interference) because it uses simple, robust modulation and coding rather than capacity-approaching codes. Equation (7) assumes an ideal band-limited additive white Gaussian noise (AWGN) channel and unlimited coding complexity; it is an upper bound, not a prediction of throughput.

6.2 Timing advance and the maximum cell radius

Problem. The BTS expects each burst to arrive inside its timeslot. A distant MS sees the downlink late and its uplink arrives late again, so it must transmit early by the round-trip delay.

Model. Line-of-sight propagation at speed c; the timing advance (TA) is signalled in steps of one bit period, $\text{TA} \in \{0, 1, \ldots, 63\}$.

Derivation. A round-trip delay of $2d/c$ must be compensated by $\mathrm{TA} \cdot T_\mathrm{b}$, so one TA step corresponds to a distance

$$\Delta d = \frac{c\,T_\mathrm{b}}{2} = \frac{2.998 \times 10^8\,\mathrm{m/s} \times 3.692\,\mu\mathrm{s}}{2} \approx 553\,\mathrm{m}, \qquad d_\mathrm{max} = 63\,\Delta d \approx 34.9\,\mathrm{km}. \tag{8}$$

Interpretation and limits. The familiar "35 km" limit of a GSM cell is a consequence of the 6-bit TA field, not of the link budget. Extended-range cells overcome it by using two adjacent timeslots per user, at the cost of capacity. The same reasoning, with a finer step, gives the LTE timing advance in Lecture 8.

6.3 Co-channel interference and the cluster size

Problem. How tightly can carriers be reused? Smaller reuse distance means more carriers per cell but stronger interference.

Model and assumptions. Hexagonal cells of radius R_c, clusters of $N = i^2 + ij + j^2$ cells (i, j non-negative integers), power-law path loss $P_\mathrm{rx} \propto d^{-n}$ with exponent n, a user at the cell edge, and only the six first-tier co-channel cells at reuse distance $D = R_\mathrm{c}\sqrt{3N}$ [9].

Derivation. The wanted signal is $C \propto R_\mathrm{c}^{-n}$ and each of the six interferers contributes approximately D^{-n}, hence

$$\frac{C}{I} \approx \frac{R_\mathrm{c}^{-n}}{6\,D^{-n}} = \frac{1}{6}\left(\frac{D}{R_\mathrm{c}}\right)^{n} = \frac{\left(\sqrt{3N}\right)^{n}}{6}. \tag{9}$$

Interpretation. Equation (9) grows with both N and n: a large cluster puts interferers further away, and a steep path loss attenuates them faster. Since GSM tolerates a C/I of roughly 9 dB, Figure 1 suggests that $N = 4$ (13.8 dB for $n = 4$) is comfortable, and $N = 3$ is feasible for $n = 4$ but marginal for $n = 3.5$. In practice operators combined tight reuse with frequency hopping and power control, which average and reduce interference.

Limitations. The model ignores shadow fading (which varies C/I by several dB), sectorisation (which removes some interferers), and the fact that not all interferers are active at once. It is a planning first estimate, not a guarantee.

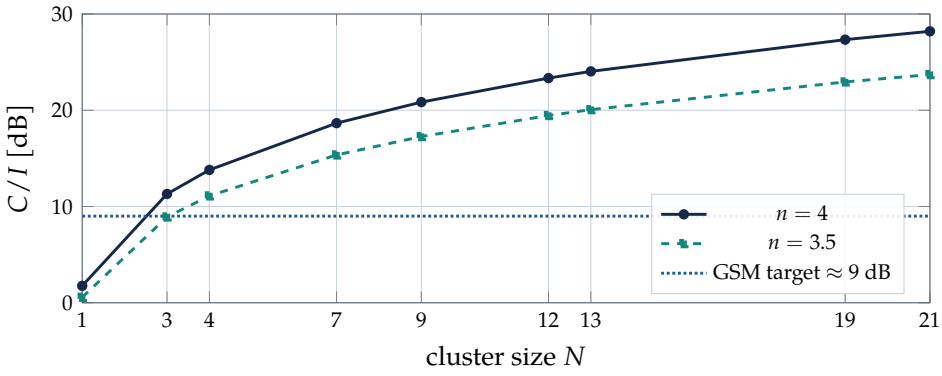


Figure 1. First-tier C/I from Eq. (9) versus cluster size for two path-loss exponents. The horizontal line marks the approximate GSM protection ratio.

6.4 Packet throughput with multiple timeslots

For a user allocated n_{TS} timeslots with per-timeslot rate r_{CS}, the peak radio-link-control (RLC) data rate is

$$R_{\mathrm{user}} = n_{\mathrm{TS}}\, r_{\mathrm{CS}}\, (1 - p_{\mathrm{BLER}}), \tag{10}$$

where p_{BLER} is the block error ratio and the factor $(1 - p_{\mathrm{BLER}})$ assumes every erroneous block is retransmitted once successfully (a simple ARQ model). Higher coding schemes increase r_{CS} but, at a given C/I, also increase p_{BLER}; the best scheme maximises the product. Figure 2 shows the peak values.

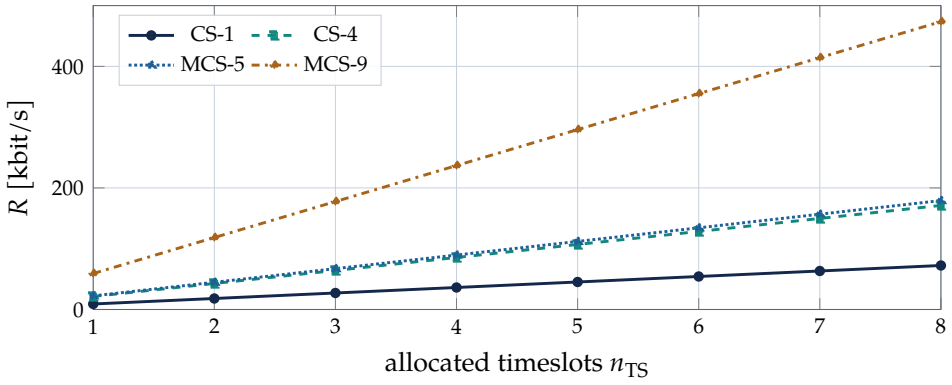


Figure 2. Peak per-user RLC data rate versus number of timeslots ($p_{\mathrm{BLER}} = 0$) for two GPRS and two EGPRS schemes. Real throughput is lower because of retransmissions, sharing among users and higher-layer overhead.

7 Network architecture and diagrams

Figure 3 combines the circuit-switched GSM network and the GPRS packet overlay. Two observations are worth making. First, the BSS is shared: the same BTS carries voice timeslots towards the MSC and packet timeslots towards the SGSN, and the BSC splits them. Second, the HLR is shared: a single subscription record serves both domains, which is why GPRS could be launched without re-provisioning subscribers.

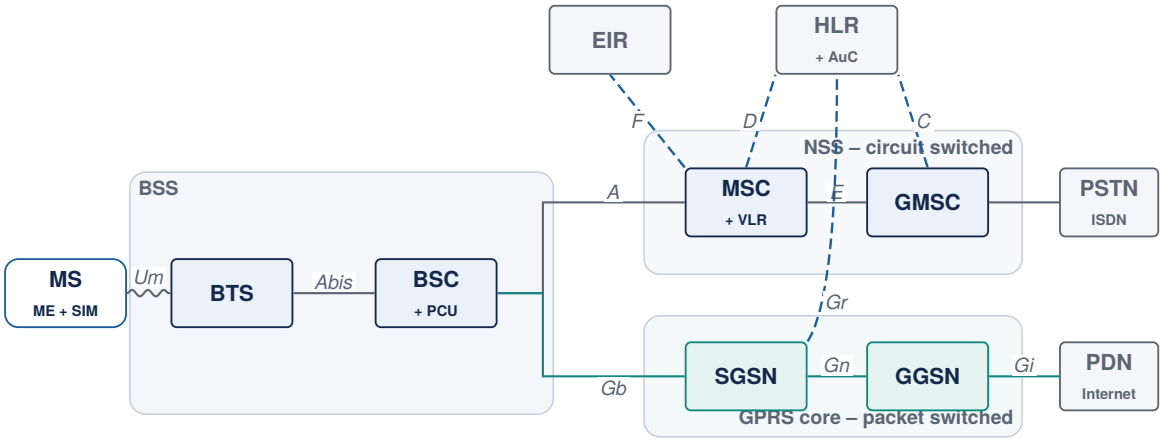


Figure 3. GSM network with the GPRS overlay. Solid grey: transport links; teal: packet-switched links; dashed blue: MAP signalling towards the HLR and EIR.

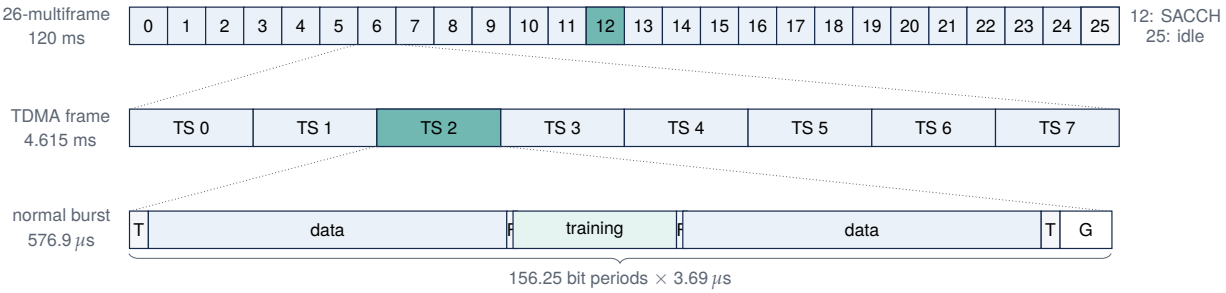


Figure 4. GSM time structure. A 26-multiframe (120 ms) contains 24 traffic frames, one SACCH frame and one idle frame; each TDMA frame contains eight timeslots; a normal burst carries 114 data bits.

8 Step-by-step procedures

Procedure 1 • Authentication and start of ciphering

1. The MS sends a Location Updating Request (or call set-up request) identifying itself with its TMSI, or with its IMSI if no TMSI is valid.
2. If the VLR has no unused triplet, it requests a batch from the HLR/AuC with MAP Send Authentication Info; the AuC returns several triplets.
3. The MSC/VLR sends Authentication Request containing only RAND.
4. The SIM computes SRES′ and K_c from Eq. (1) and the MS returns SRES′.
5. The VLR compares SRES′ with SRES. On a match it orders ciphering by sending K_c to the BSS in a BSSMAP Cipher Mode Command.
6. The BTS sends Ciphering Mode Command; the MS starts A5 with K_c and answers Ciphering Mode Complete. From now on Um is encrypted.
7. The VLR may allocate a new TMSI, sent encrypted, so the IMSI is protected next time.

Procedure 2 • GPRS attach and PDP context activation

1. **GPRS attach.** The MS sends Attach Request to the SGSN. The SGSN authenticates the MS (triplets from the HLR over Gr), registers itself in the HLR with Update GPRS Location, receives the subscription with Insert Subscriber Data and returns Attach Accept with a P-TMSI. The MS is now in READY state but has no IP address.
2. **Session request.** The MS sends Activate PDP Context Request with an NSAPI (identifier of the context), the requested access point name (APN) and QoS.
3. **Gateway selection.** The SGSN checks the subscription and resolves the APN with DNS to the address of a GGSN.
4. **Tunnel set-up.** The SGSN sends GTP-C Create PDP Context Request; the GGSN allocates an IP address (or obtains one from a RADIUS/DHCP server) and returns its tunnel end-point identifier (TEID).
5. **Completion.** The SGSN sends Activate PDP Context Accept with the IP address. User packets now flow: MS–SGSN over LLC/SNDCP, SGSN–GGSN inside a GTP-U tunnel

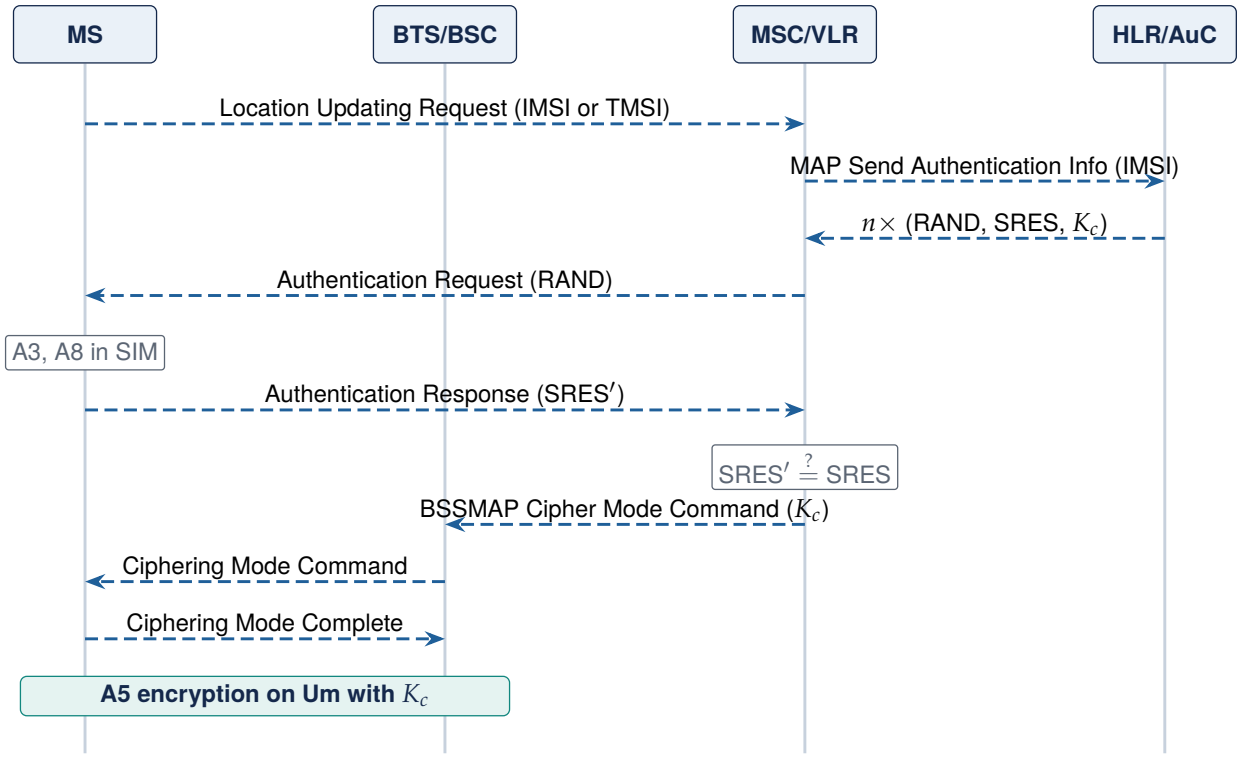


Figure 5. Message sequence of GSM authentication and ciphering (Procedure 1).

> identified by TEIDs.

The procedure illustrates a principle that the course will meet again in LTE (the default bearer, Lecture 9) and 5G (the PDU session, Lecture 20): *being attached* (known to the network) and *having connectivity* (an IP address and a tunnel) are separate states, established by separate procedures. LTE later merged them for the first PDN connection to reduce delay.

9 Worked examples

Worked example 1 • Capacity of a GSM-900 operator

Problem. An operator owns 10 MHz of paired GSM-900 spectrum and plans with cluster size $N = 4$. How many full-rate voice channels does a cell offer?

Solution. Number of carriers: $\lfloor 10\,\text{MHz}/200\,\text{kHz} \rfloor = 50$; one is kept as a guard, so 49 are usable. Per cell: $\lfloor 49/4 \rfloor = 12$ carriers, i.e. $12 \times 8 = 96$ timeslots. Reserving one timeslot for the broadcast/common control channel and one for SDCCH signalling leaves about 94 traffic channels.

Interpretation. 94 simultaneous calls per cell is the hard limit of circuit switching. How much *traffic* (in erlangs) these channels carry at a given blocking probability is computed with the Erlang-B formula in Lecture 2.

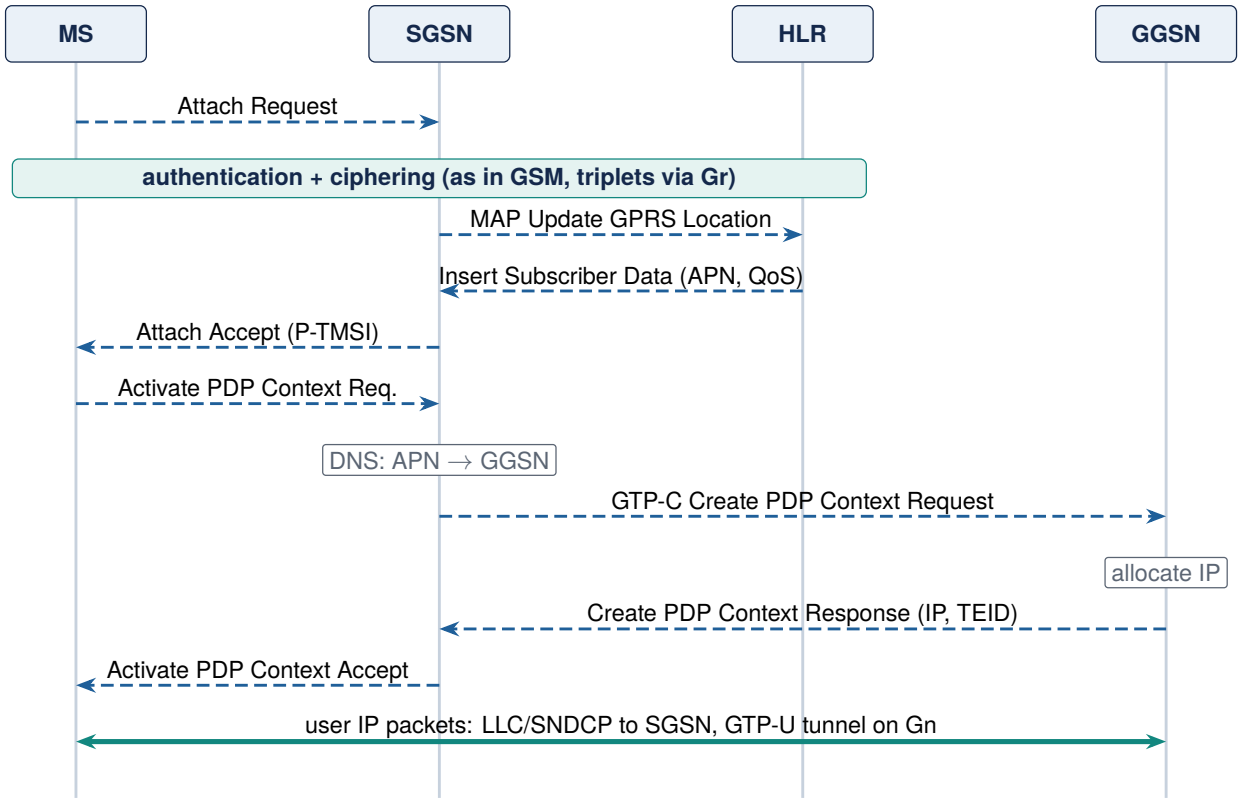


Figure 6. GPRS attach followed by PDP context activation (Procedure 2). Dashed blue: signalling; solid teal: user plane.

Worked example 2 • Timing advance

Problem. The BTS measures that an MS's bursts arrive 55 µs late. Which TA value is signalled and how far is the MS?

Solution. The round trip lasts 55 µs, so TA $= \text{round}(55/3.692) = 15$ and $d \approx 15 \times 553\,\text{m} \approx 8.3\,\text{km}$. (Directly: $d = c\,\tau/2 = 3 \times 10^8 \times 55 \times 10^{-6}/2 = 8.25\,\text{km}$.)

Interpretation. The TA quantisation error is up to half a step, ≈ 277 m; TA therefore also gives the network a coarse estimate of the user's distance, the basis of early cell-ID+TA positioning.

Worked example 3 • Choosing a GPRS coding scheme

Problem. At a given C/I a user observes $p_{\text{BLER}} = 0.05$ with CS-2 and $p_{\text{BLER}} = 0.45$ with CS-4. With four timeslots, which scheme gives the higher throughput according to Eq. (10)?

Solution.

$$R_{\text{CS-2}} = 4 \times 13.4 \times (1 - 0.05) = 50.9\,\text{kbit/s},$$
$$R_{\text{CS-4}} = 4 \times 21.4 \times (1 - 0.45) = 47.1\,\text{kbit/s}.$$

Interpretation. CS-2 wins despite its lower nominal rate. Link adaptation is precisely the selection of the scheme that maximises this product; the same logic reappears with CQI and MCS tables in LTE (Lecture 7).

10 Technology comparison

Table 2. GSM, GPRS and EDGE compared. Rates are peak RLC rates per timeslot or per user with eight timeslots; achievable user rates are lower.

	GSM (circuit data)	GPRS	EDGE (EGPRS)
Switching	Circuit	Packet	Packet
Modulation	GMSK	GMSK	GMSK and 8-PSK
Bits per symbol	1	1	1 or 3
Rate per timeslot	9.6 or 14.4 kbit/s	9.05–21.4 kbit/s	8.8–59.2 kbit/s
Peak (8 TS)	–	171.2 kbit/s	473.6 kbit/s
Resource use	Reserved for the call	Shared, on demand	Shared, on demand
Retransmission	None (FEC only)	ARQ in RLC	ARQ + incremental redundancy
Core nodes	MSC, GMSC	SGSN, GGSN	SGSN, GGSN (unchanged)
Charging basis	Time	Volume	Volume

Trade-offs. GPRS made data cheap (volume-based charging, no idle circuit) but introduced queuing delay and variable throughput; round-trip times of several hundred milliseconds were typical. EDGE tripled the peak rate with a software and transceiver upgrade, but 8-PSK is not constant-envelope, so it needs more linear power amplifiers, and its high schemes are only usable close to the BTS. Neither changed the fundamental limit: data rides on 200 kHz carriers designed for voice. Breaking that limit required a new radio interface, WCDMA in 3G (Lecture 3).

11 Key takeaways

Key takeaways

- GSM is a *system* standard: open interfaces (A, Um) and a separable identity (SIM/IMSI) enabled multi-vendor networks and roaming.
- All GSM timing follows from $R = 270.833\,\text{kbit/s}$: bit 3.69 µs, slot 577 µs, frame 4.615 ms, 26-multiframe 120 ms; a full-rate TCH carries 22.8 kbit/s gross.
- The 6-bit timing advance limits a normal cell to about 35 km.
- Authentication uses triplets (RAND, SRES, K_c); it is one-way, and encryption stops at the BTS.
- GPRS adds PCU, SGSN and GGSN; the PDP context and the GTP tunnel are the ancestors of the LTE bearer and the 5G PDU session.
- EDGE raises the bit rate per symbol (8-PSK) and adds incremental redundancy, an early hybrid ARQ.
- The first-tier model $C/I \approx (\sqrt{3N})^n/6$ links reuse, path loss and interference; it is a first planning estimate.

12 Exercises

Exercise 1 • Level 1 – conceptual • Architecture

Explain why the HLR stores the address of the VLR but not the cell in which the subscriber is located. What would be the signalling consequence of storing the cell?

Exercise 2 • Level 1 – conceptual • Security

List two attacks made possible by the one-way authentication of GSM and say which later mechanism (Lecture 11) prevents each.

Exercise 3 • Level 2 – numerical • Half-rate channel

A half-rate traffic channel uses a timeslot in every second traffic frame. Using Eq. (6), compute its gross bit rate, and explain how it doubles the voice capacity of the cell in Example 1.

Exercise 4 • Level 2 – numerical • Reuse planning

Using Eq. (9), find the smallest valid cluster size N that guarantees $C/I \geq 12\,\text{dB}$ for (a) $n = 4$ and (b) $n = 3$.

Exercise 5 • Level 2 – numerical • EDGE throughput

A user has five timeslots. With MCS-9 the block error ratio is 0.30; with MCS-7 (44.8 kbit/s per timeslot) it is 0.08. Which scheme should link adaptation choose, and what is the expected throughput?

Exercise 6 • Level 3 – analytical • Overhead budget

Starting from Eqs. (2) and (4), compute what fraction of the gross bit rate of one timeslot is used by (i) guard period, (ii) tail bits and stealing flags, (iii) training sequence, (iv) SACCH and idle frames. Compare the total with the cyclic-prefix overhead of LTE ($\approx 7\%$) and discuss why the comparison is not entirely fair.

Exercise 7 • Level 3 – analytical • Generalising the C/I model

Extend Eq. (9) to a cell divided into three 120° sectors, where only two first-tier interferers fall into the main lobe of the sector antenna. Derive the new expression and the gain in dB, and discuss which assumptions of the original model the result inherits.

Solutions

Solution to Exercise 1

The cell changes every few minutes for a moving user; if the HLR tracked it, every cell change of every subscriber in the world would generate signalling to the home network, often internationally. Storing only the VLR address limits HLR updates to changes of MSC area, and the location area kept in the VLR limits radio paging to a group of cells. This two-level hierarchy

(HLR → VLR → LA) is the core of GSM mobility management.

Solution to Exercise 2

(1) A false base station can request the IMSI in clear and track the user (IMSI catching); 5G conceals the permanent identity with SUCI, and LTE/UMTS reduce the exposure. (2) A false base station can accept the MS without authentication and order no ciphering or a weak cipher, then relay traffic (man-in-the-middle). UMTS/LTE mutual authentication with the AUTN token (Lecture 11) lets the USIM verify that the challenge was produced by its home network, and integrity protection of signalling prevents tampering with the security-mode command.

Solution to Exercise 3

A half-rate channel gets 12 of the 24 traffic frames per 26-multiframe, so $R_{\mathrm{TCH/H}} = 114 \times 12/0.120 = 11.4\,\mathrm{kbit/s}$. Two half-rate users share one timeslot, so the 94 timeslots of Example 1 can carry up to 188 calls, at the price of a lower-rate (lower-quality) speech codec.

Solution to Exercise 4

We need $(\sqrt{3N})^n/6 \geq 10^{1.2} = 15.85$, i.e. $(3N)^{n/2} \geq 95.1$. (a) $n = 4$: $(3N)^2 \geq 95.1 \Rightarrow N \geq 3.25$; the smallest valid N is 4 ($C/I = 13.8\,\mathrm{dB}$). (b) $n = 3$: $(3N)^{1.5} \geq 95.1 \Rightarrow 3N \geq 20.8 \Rightarrow N \geq 6.94$; the smallest valid N is 7 ($C/I = 10\log_{10}(21^{1.5}/6) = 12.05\,\mathrm{dB}$). Valid sizes are $N = i^2 + ij + j^2 \in \{1, 3, 4, 7, 9, 12, \dots\}$.

Solution to Exercise 5

$R_{\text{MCS-9}} = 5 \times 59.2 \times 0.70 = 207.2\,\mathrm{kbit/s}$ and $R_{\text{MCS-7}} = 5 \times 44.8 \times 0.92 = 206.1\,\mathrm{kbit/s}$. The two are practically equal; MCS-9 is marginally better on throughput, but MCS-7 gives fewer retransmissions and therefore lower delay and jitter. With incremental redundancy the effective MCS-9 throughput would be higher than the simple model predicts, because failed blocks still contribute soft information.

Solution to Exercise 6

Per 156.25-bit slot: guard $8.25/156.25 = 5.3\%$; tail and flags $(6+2)/156.25 = 5.1\%$; training $26/156.25 = 16.6\%$; data $114/156.25 = 73.0\%$. Of the 26 frames, 2 do not carry traffic, so the useful fraction is $0.730 \times 24/26 = 67.4\%$: $270.833/8 \times 0.674 = 22.8\,\mathrm{kbit/s}$, consistent with Eq. (6). Total overhead $\approx 33\%$. LTE's cyclic prefix costs about 7%, but LTE also spends resources on reference signals, control region and guard bands; a fair comparison must count all of them (Lecture 7).

Solution to Exercise 7

With two interferers, $C/I \approx (\sqrt{3N})^n/2$, which is three times (4.8 dB) better than Eq. (9). For $N = 4$, $n = 4$: $10\log_{10}(144/2) = 18.6\,\mathrm{dB}$. The result still assumes ideal hexagons, equal powers, no shadowing and a worst-case user at the cell edge; in addition it assumes ideal antenna patterns with no back-lobe leakage. Sectorisation trades trunking efficiency (each sector has fewer channels) for interference reduction.

13 Further reading

The architecture of all 3GPP systems is summarised in [5]; the GSM radio structure is specified in [2] and security functions in [7]. GPRS stage-2 procedures are in [1] and layer-3 messages in [4]. The textbook by Eberspächer et al. [8] remains the most complete single description of GSM and GPRS, and Rappaport [9] derives the cellular reuse model used in Section 6.3. MAP is specified in [3]; identities in [6].

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